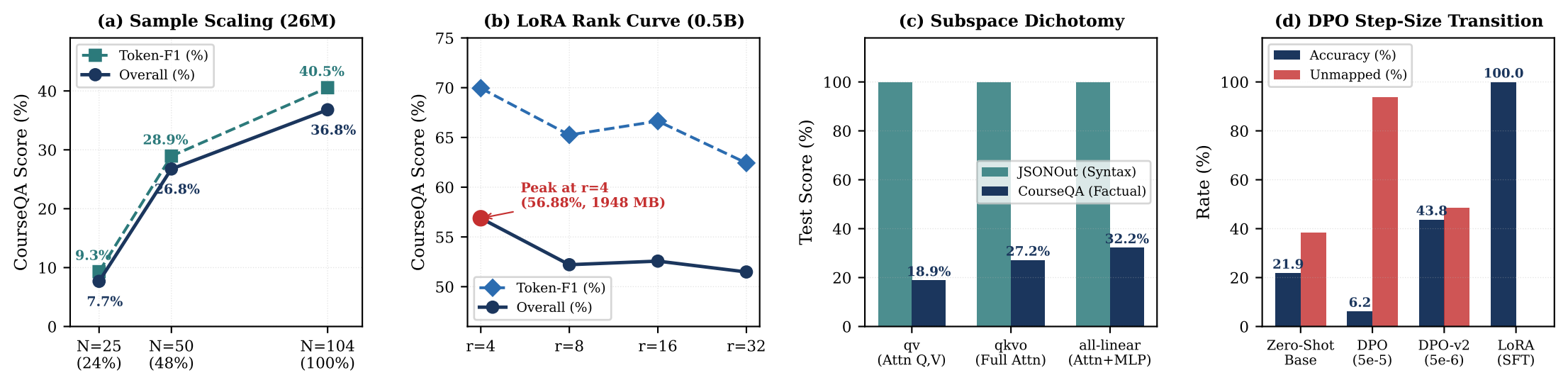
(a) Sample Scaling (26M)
Token-F1 (%)
Overall (%)
CourseQA Score (%)
40.5%
28.9%
36.8%
26.8%
9.3%
7.7%
N=25 (24%)
N=50 (48%)
N=104 (100%)
(b) LoRA Rank Curve (0.5B)
CourseQA Score (%)
Peak at r=4
(56.88%, 1948 MB)
Token-F1 (%)
Overall (%)
r=4
r=8
r=16
r=32
(c) Subspace Dichotomy
Test Score (%)
JSONOut (Syntax)
CourseQA (Factual)
18.9%
27.2%
32.2%
qv (Attn Q,V)
qkvo (Full Attn)
all-linear (Attn+MLP)
(d) DPO Step-Size Transition
Accuracy (%)
Unmapped (%)
Rate (%)
21.9
6.2
43.8
100.0
Zero-Shot Base
DPO (5e-5)
DPO-v2 (5e-6)
LoRA (SFT)